

ENTERPRISE AI KNOWLEDGE ASSISTANT

Enterprise Project Management Toolkit

Test Plan & UAT Sign-Off

Quality Assurance Strategy – Enterprise AI Knowledge Assistant

Document Title	Test Plan & UAT Sign-Off
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1. Purpose

This Test Plan defines the testing strategy, scope, and acceptance criteria for the Enterprise AI Knowledge Assistant (EAIKA), covering functional testing of the RAG pipeline, non-functional testing of performance and security, and formal User Acceptance Testing (UAT) with the pilot department.

2. Test Scope

2.1 In Scope

- Document ingestion across all supported formats (PDF, DOCX, PPTX, XLSX, TXT, CSV, EML, PNG, JPG, JPEG).
- OCR text extraction accuracy for image-based and scanned content.
- Semantic retrieval accuracy, including queries phrased differently from source document wording.
- Answer generation quality and correctness of source citations.
- Administrator Dashboard functions: upload, build, rebuild, statistics, inventory.
- Non-functional: response latency, concurrent user handling, and security controls.

2.2 Out of Scope

- Testing of domains beyond the Phase 1 Project & Program Management pilot (e.g., HR, Finance, Healthcare).
- Load testing at multi-department, organization-wide scale (deferred to future phase readiness testing).

3. Test Levels & Approach

Test Level	Objective	Owner
Unit Testing	Validate individual functions in the services layer (chunking, embedding calls, retrieval queries) in isolation.	Engineering Team
Integration Testing	Validate the end-to-end pipeline: ingestion → embedding → ChromaDB → retrieval → LangChain → Ollama/Qwen2.5 → UI.	Engineering Team / QA
System Testing	Validate the application as a whole against functional requirements (FRD) using representative pilot documents.	QA Team
Non-Functional Testing	Validate response latency, concurrency handling, and resilience to malformed documents.	QA Team / Infrastructure
Security Testing	Validate access restrictions on administrator functions and confirm no external data egress occurs.	Information Security
User Acceptance Testing (UAT)	Validate the system meets real business needs from the perspective of pilot department users.	Pilot Department / Product Owner

4. Sample Functional Test Cases

Test ID	Test Description	Expected Result
TC-01	Upload a PDF project status report and trigger a knowledge base build.	Document appears in inventory; build completes; chunk count increases; document becomes query able.
TC-02	Ask “What is the current project health?” after ingesting a status report using different terminology (e.g., “on track” vs. “healthy”).	System returns a correct, source-cited answer despite the terminology difference (semantic search validation).

TC-03	Ask a question with no relevant content in the knowledge base.	System responds that no relevant information was found rather than fabricating an answer.
TC-04	Upload a scanned (image-based) PDF containing a RAID log.	OCR extracts readable text; content becomes searchable and query able.
TC-05	Ask “What are the highest project risks?” after ingesting a RAID Log.	Answer correctly reflects top risks with citation to the RAID Log document.
TC-06	Administrator triggers a full knowledge base rebuild.	Rebuild completes successfully; statistics reflect updated document/chunk counts; no data loss.
TC-07	Non-administrator user attempts to access the Administrator Dashboard upload function.	Access is denied or the function is not exposed, per configured access restrictions.
TC-08	Submit a query while the Ollama service is intentionally stopped.	System surfaces a clear timeout/error message rather than hanging indefinitely or crashing.

5. Non-Functional Test Criteria

Criterion	Target
Response latency (typical query)	Within the target defined in the Enterprise Readiness & SLA document for the deployed hardware profile.
Concurrent pilot users supported	At least the expected pilot department concurrency without material latency degradation.
Data egress	Zero document content or query text transmitted to any external/cloud AI service, verified via network traffic review.
Ingestion failure handling	Corrupted or unsupported files are rejected with a clear administrator-facing error, with no impact on the rest of the knowledge base.

6. Defect Severity Classification

Severity	Definition	Example
Critical	System unusable or produces materially incorrect information without any indication of failure.	Answers are generated with fabricated citations pointing to non-existent documents.
High	Major function impaired; workaround difficult or unavailable.	Knowledge base rebuild fails silently and stale answers are served.
Medium	Function impaired but workaround available.	OCR misreads a subset of scanned documents; can be manually re-uploaded as text.
Low	Minor / cosmetic issue with no functional impact.	Formatting inconsistency in the citation display.

7. UAT Approach

UAT is conducted with representative users from the pilot PMO / project delivery department over a two-week window. Users execute a defined set of realistic questions against the live pilot knowledge base and rate each answer for accuracy, relevance, and citation correctness. A minimum 80% accuracy threshold across the UAT test set, as defined in the PRD, is required for sign-off.

8. UAT Sign-Off

UAT Criterion	Result	Accepted (Y/N)
Minimum 80% answer accuracy achieved across UAT test question set		
All Critical and High severity defects resolved or formally accepted with mitigation		
Source citation present and correct for all sampled answers		
Administrator Dashboard functions validated by pilot administrators		
Performance within Enterprise Readiness & SLA targets during UAT		

9. Approval

Name / Role	Signature	Date
QA / Test Lead		
Product Owner		
Pilot Department Representative		